An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning

Implementation and evaluation of Krishnan & Naik (2025)

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CS 552 Group Project

ASSIGNED PAPER

Sruthi Krishnan and Manohar Naik S. "An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning." *Security and Privacy*, 8(4):e70057, 2025.

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JOURNAL

Wiley · Security & Privacy

Roadmap: from “why?” to “does it work?”

1 · Problem	2 · Idea	3 · How	4 · Build	5 · Results
Phishing links & concept drift	Watch drift, re-pick features, learn online	Features → LASSO+MI → SGD → KS → loop	`drift_phish`, data, 5 experiments	What reproduced — and what did not

A phishing detector that **notices when attackers change style**, re-picks the most useful clues, and **keeps learning in milliseconds** instead of retraining from scratch.

Phishing starts with a link that looks almost right

Real

`https://www.paypal.com/signin`

Phishing

`http://secure-paypal-login.tk/confirm/...`

Why this is hard

Blocklists lag brand-new links. We need a model that judges a URL **it has never seen** — from the URL string alone.

	Page content	**URL-only**
Needs	download page	URL text
Speed	slow	instant
Privacy	reads page	no page read
Safety	visits site	never visits

Key question: can URL text alone stay accurate as attacks evolve?

The catch: a model trained once goes stale (concept drift)

What is concept drift?

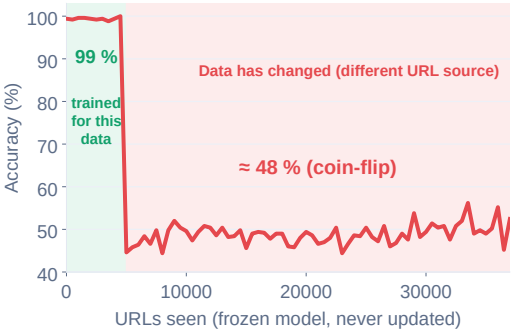
Statistics of phishing URLs **change over time** — shorter hostnames, more random characters, fewer bait words. Yesterday's model errs on today's URLs.

We measured it

34 of 58 features shift (KS, δ=0.05) between old corpora and PhishTank Jun 2024–Jan 2025.

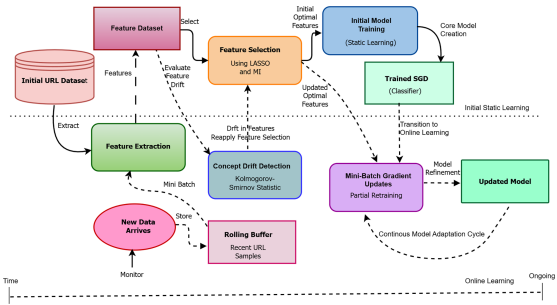
Obvious fix is expensive

Full retrain ≈ **4 s** on 210k URLs vs **1.5 ms** incremental update.



S2: frozen model scores **99%** on its train domain, then ≈**48%** (coin-flip) after the URL source changes.

Paper idea: three ingredients that stay current



Paper Fig. 1: offline once (top); online forever (bottom).

1. **Pick best clues** — LASSO + mutual information; keep top quartile F^*.
2. **Watch for change** — Kolmogorov–Smirnov test per feature vs. history.
3. **Learn on the fly** — online SGD mini-batches; no full retrain.

Paper headline claim

99.85% accuracy · **0.067%** FPR

Our task: implement every box of Fig. 1

Assignment

Implement the paper's methodology and reproduce experiments. Package `drift_phish` (∼1,200 LOC) + 5 experiment scripts + 16 unit tests + browser demo.

Paper box	Module	Key knobs
URL → features	`features/extractor.py`	58 lexical features
Feature selection	`selection/dynamic.py`	$S{=}w{+}I$, $\tau{=}P_{75}$
SGD classifier	`learning/online_sgd.py`	$\eta_0{=}0.01$, balanced
Drift detection	`drift/ks_monitor.py`	$\delta{=}0.05$, buf 1500
Online loop	`framework.py`	$B{=}500$, adaptive mode

Modes for fair comparison: **adaptive** (proposed) · **online_static** · **frozen** · **periodic_batch**. One command: `make all`.

Step 1 — Turning a URL into numbers

`http://secure-paypal-login.tk/confirm/account?user=1&id=88`

Feature	wikipedia.org	phishing URL
`url_length`	39	**58**
`num_hyphens`	0	**2**
`number_count`	0	**3**
`entropy`	3.78	**4.62**
`has_https` / `scheme_http`	1 / 0	**0 / 1**
`suspicious_tld`	0	**1**
`suspicious_term_frequency`	0	**4**

Result

Each URL → **58 numbers**. Pure string processing ≈ **0.16 ms**/URL — no download.

58 URL clues in six families (paper Table 3)

Length & size · 8

url / domain / path / query length, tokens, . . .

Structure & symbols · 18

dots, slashes, hyphens, `=`, `&`, `@`, `%`

Obfuscation · 9

IP host, shorteners, suspicious TLD, homoglyphs

Digits & randomness · 6

number_count, digit_ratio, entropy, host_entropy

Bait & brands · 11

login, verify, secure, free, brand proximity

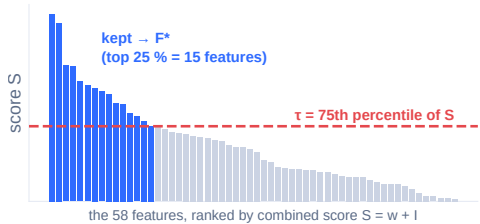
Scheme & redirects · 6

https/http; redirect & domain-mismatch = **lexical proxies**

Honest note

Paper's redirect-chain / domain-mismatch need live HTTP. We use lexical proxies (e.g. contains `redirect=` or a second `http`) so everything stays URL-string-only.

Step 2 — Not every clue helps: keep the top quarter



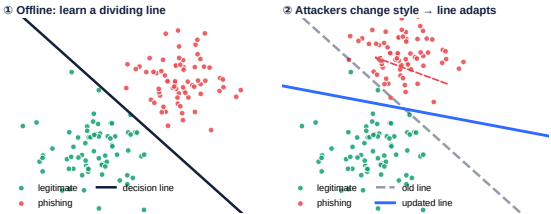
Rank 58 features by S; top quartile (blue) = active set F^*.

1. **LASSO** (λ=10^{-4}) → weights w (useless → 0)
2. **Mutual information** → scores I (also non-linear clues)
3. Combined score $S = w + I$
4. Keep $S \geq \tau$, $\tau = P_{75}(S)$ ⇒ $|F^*|$=15

Why shrink & re-select?

Faster, less overfitting, interpretable. Best clues **change** when attackers change tactics.

Step 3 — Linear SVM + SGD: draw a line, then nudge it



Offline: separate legit vs phishing. Online: mini-batch updates tilt the line.

Why SGD?

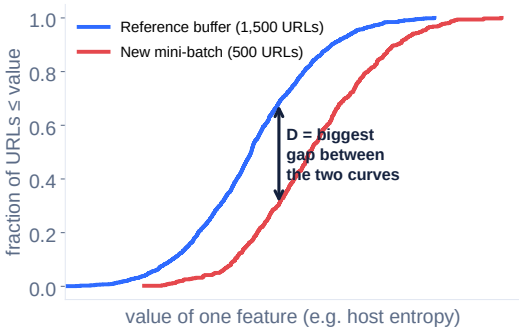
Learns from **small chunks** — `partial_fit` on a 500-URL batch instead of re-reading the whole corpus.

Variants: **SGD-SVM** **SGD-LR** **SGD-Perceptron**

Table 4 knobs we used

η_0=0.01 · adaptive LR · balanced · early stop · buffer 1500 · B=500 · KS δ=0.05 · 80:20 · seed 42

Step 4 — KS test: “do these two samples look alike?”



D = largest vertical gap between reference CDF and new-batch CDF.

1. Reference = last **1,500** labelled URLs
2. Every mini-batch B=500: KS per feature
3. $p < \delta$=0.05 $\Rightarrow$ that feature drifted

Paper leaves the event rule open

Our rule: drift event when $\geq$**25%** of active features F^* fail KS.

Step 5 — The online loop (every mini-batch)

500 URLs → 1 Predict → 2 KS test → 3 Drift? → 4 Learn (partial_fit)

If drift is detected

1. Mark drifted features in F^* obsolete
2. Re-run LASSO+MI on the buffer
3. Blend with importance history H (50/50)
4. New F^* = top quartile again
5. Mask removed features; 3 replay epochs
6. Reset KS reference to the new buffer

Why predict first?

Prequential evaluation: every URL scored by a model that has **not** seen it yet.

Data and the five experiments

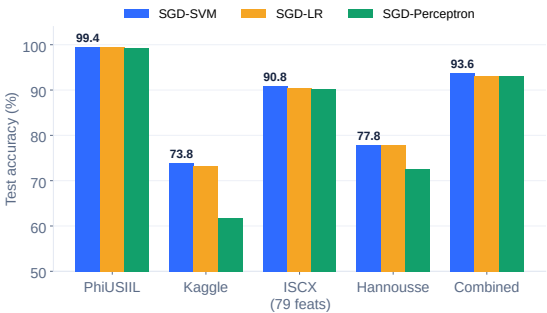
Corpus	URLs	Role
PhiUSIIL	235k	Table 7
Kaggle subset	21k	Table 6
ISCX-URL-2016	15.4k	Table 5 (79 feats)
Hannousse 2021	11.4k	raw URLs (ISCX stand-in)
Combined	263k	Fig. 6
PhishTank stream	12,888	Jun 24–Jan 25

1. Offline (T5–7, Fig. 6)
2. Online streams S1 / S2
3. Ablation (Table 9)
4. Runtime (T10–11)
5. Drift analysis (Figs. 4,5,7; T8)

Setup

Positive = phishing · 80:20 seed 42 · streams: 5 seeds (mean±std) · B=500.

Results 1 — Offline: SGD-SVM best linear model



Our Fig. 6 analogue: SVM best or tied on every corpus.

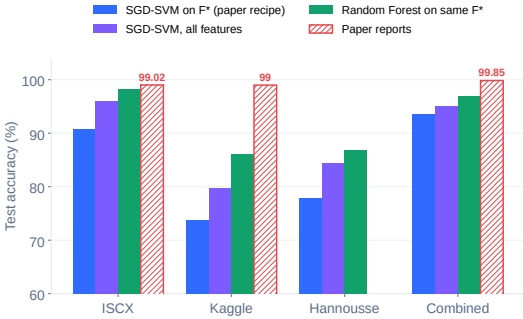
Corpus	Ours	Paper	
PhiUSIIL	**99.38**	98.78	✓
ISCX	90.83	99.02	≈
Kaggle	73.81	99.00	≈
Combined	93.58	99.85	≈

PhiUSIIL highlight

99.38% acc · FPR **0.004%**

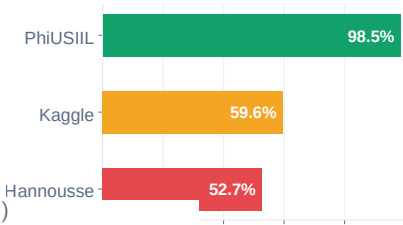
≈ = same method ordering, lower absolute level.

Why Kaggle / ISCX / combined numbers are lower



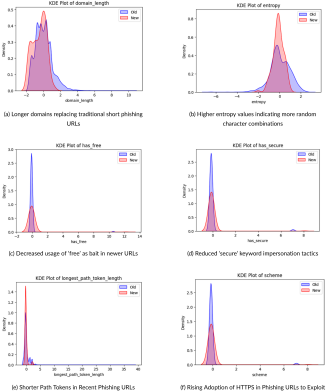
All-features lifts ISCX 90.8→96.1%; RF on same F^* reaches 98.3%.

1. **Feature set:** only top-quartile F^*; paper does not publish exact defs.
2. **Data versions:** no published cleaning / versions; ISCX only as vectors.
3. **Corpora differ:** PhiUSIIL benign mostly bare domains (easy); Kaggle/Hannousse much harder.



Results 2 — Drift is real: old vs new phishing

Paper Fig. 4



(a) Longer domains replacing traditional short phishing URLs

(b) Higher entropy values indicating more random character combinations

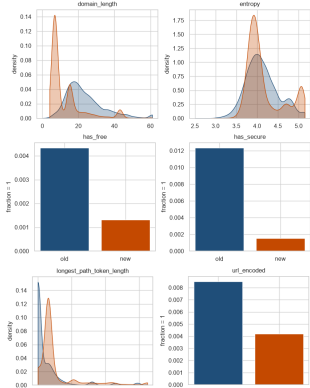
(c) Decreased usage of 'free' as bait in newer URLs

(d) Reduced 'secure' keyword impersonation tactics

(e) Shorter Path Tokens in Recent Phishing URLs Mimicking Legitimate Site Structures

(f) Rising Adoption of HTTPS in Phishing URLs to Exploit User Trust Indicators

Ours (same features)

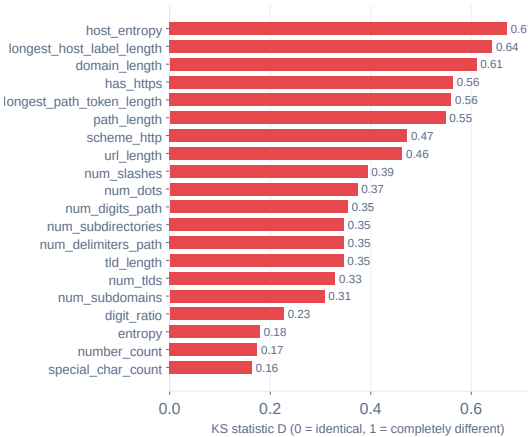


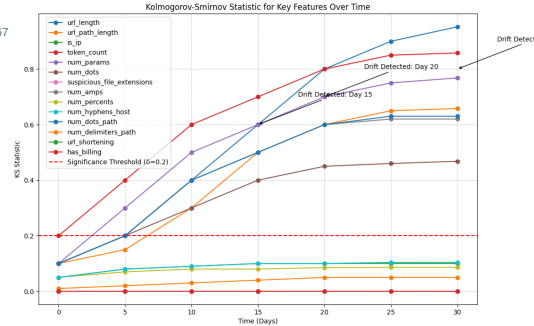
34 / 58

features differ (KS, δ=0.05)

New URLs: narrower ranges; bait words rarer. Paper premise holds on Jun 2024–Jan 2025 data.

KS statistic per feature (paper Fig. 7)





Reading it

Longer bar = bigger shift. Biggest: host entropy, host/domain length, https/http, path & URL length — length/structure family, as in the paper.

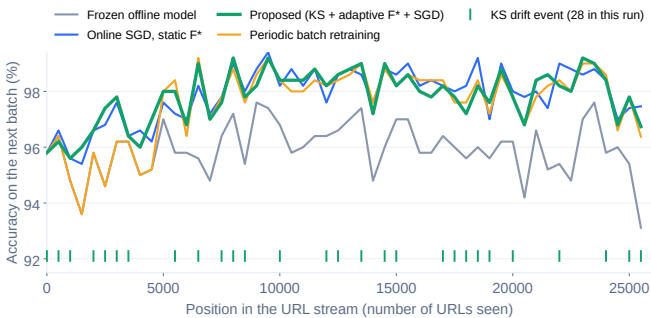
Selected features change after drift (paper Table 8)

Initial	Stage 1	Stage 2	Stage 3
length, structure, https/http, digit_ratio, ...	+entropy, host_entropy, longest_host_label −has_https, num_dots, scheme_http	+num_ampersand −num_hyphens	+num_digits_query −num_ampersand

What this shows

Feature set moves toward **host-name randomness** and **query-string** clues — matching the paper's story that newer phishing relies on query obfuscation. **S1: 28.6 drift events** **S2: 7 events**

Results 3 — Stream S1: online beats frozen by +5 pts



Prequential accuracy, 25,776 URLs (PhishTank window). Green ticks = KS events.

Frozen
92.63%

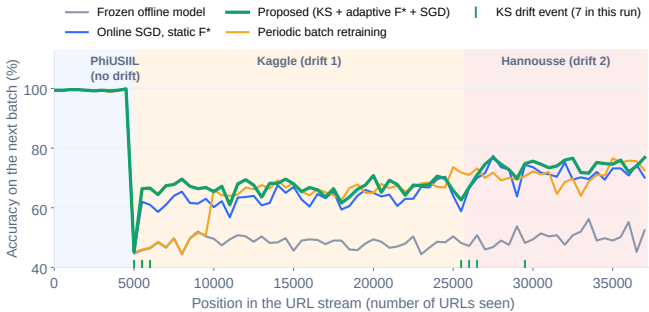
Online static F^*
97.77%

Proposed adaptive
97.80%

Periodic batch
97.31%

+5.1 pts over frozen, much lower variance (± 0.05 vs ± 6.4). Adaptive $\approx$ static here — shift is gentle.

Stream S2 — under big shifts, adaptive selection helps

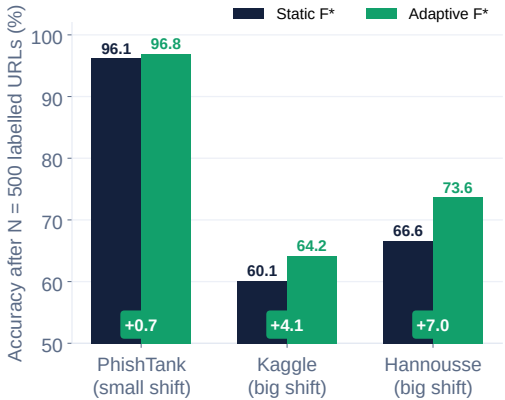


PhiUSIIL → Kaggle → Hannousse. Shading = corpus; ticks = drift events.

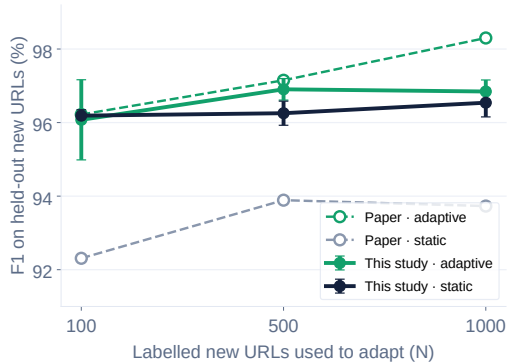
Frozen	55.5%
Online static	70.5%
Proposed adaptive	73.2%
Periodic batch	70.0%

Adaptive gains **+2.7 pts** over static online — feature re-selection matters when the distribution jumps hard.

Ablation (Table 9) — when adaptive F^* helps



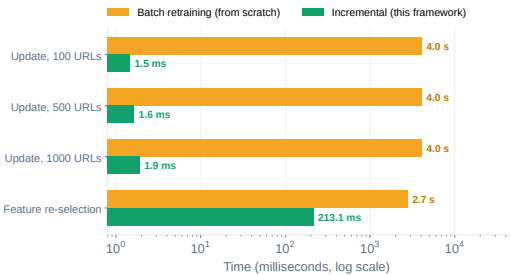
Accuracy vs. # new labelled URLs N.



F1; gains clearest on hard targets (Kaggle / Hannousse).

PhishTank (mild): adaptive ≈ static at high accuracy. Hard shift: adaptive lifts accuracy and lowers FPR for $N \geq 500$. Caveat: our static baseline is stronger than the paper's.

Runtime — real-time scoring, cheap updates



Per-URL latency

0.16 ms avg (1 CPU thread)

Paper: 12.4–15.2 ms (different HW)

Op	Incr.	Batch	Gain
Update 100	1.5 ms	4.0 s	**2770**×
Update 500	1.6 ms	4.0 s	2477×
Update 1k	1.9 ms	4.0 s	2140×
Feat. adapt	213 ms	2.7 s	13×

Scorecard — what reproduced vs. what is lower-level

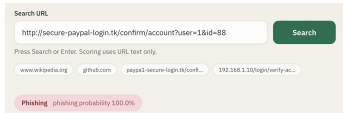
Claim / experiment	Ours	Verdict
PhiUSIIL SGD-SVM accuracy	99.38% (paper 98.78)	**matches**
SVM best of 3 SGD variants	yes	**matches**
Old vs new distributions drift	34/58 features	**matches**
F^* changes after drift (T8)	logged swaps	**matches**
Online $>$ frozen (streams)	$+5.1$ / $+15.0$ pts	**matches**
Ablation adaptive F1 (PhishTank)	$\approx$ paper	**close**
ISCX / Kaggle / Combined absolute	lower	**ordering OK**
Incremental $\gg$ batch cost	$10^3\times$	**same conclusion**

We reproduce the **method behaviour** and qualitative claims. Absolute levels on some corpora depend on unpublished feature defs / data versions.

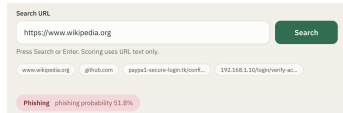
Demo — PhishCheck (browser, lexical SGD-SVM)



Clean legit URL



Obvious phishing URL



Bare domain near boundary

Limitation (be upfront)

URL-only models can false-alarm on bare domains with little context (e.g. `wikipedia.org` near the decision boundary). Live demo: `Source_Code/web/`.

Takeaways

What we learned

- URL lexical features + online SGD work in real time
- Concept drift is measurable (34/58 features)
- Online learning recovers +5 to +15 pts vs frozen
- Adaptive F^* helps most under **strong** shifts

Limitations & next

- Absolute levels need paper's exact features / cleaning
- No HTML / third-party signals (by design)
- Drift-event rule (25%) is our design choice
- Future: richer features, better bare-domain handling

Reproduce everything: `cd Source_Code && make all` · Report: `Project_Report.pdf` · Results: `Experimental_Results/RESULTS.md`

Questions?

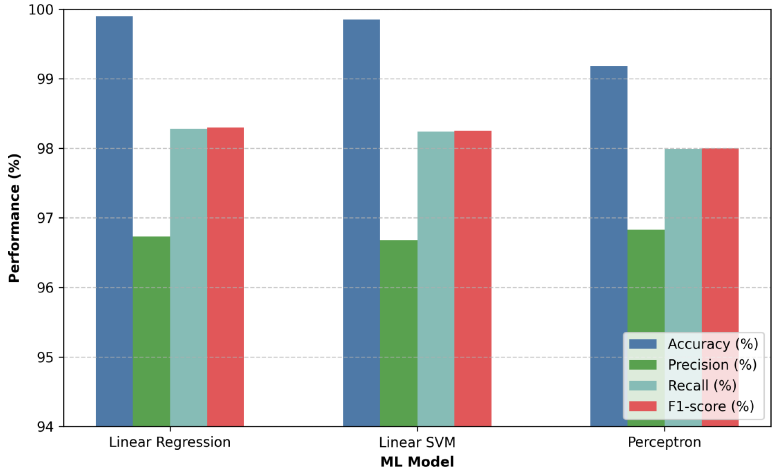
Thank you

Pratham Garg · Sumit Sharma · Vivaan Garg

CS 552 · Krishnan & Naik, *Security and Privacy* 2025

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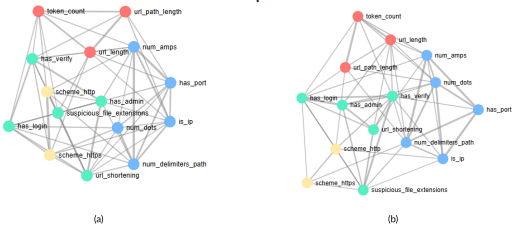
Appendix A — Combined SGD-SVM vs paper Fig. 6



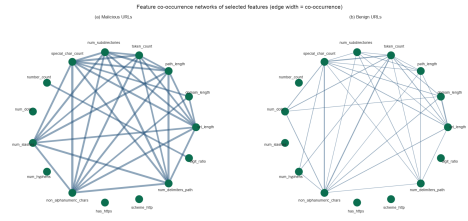
Paper Fig. 6. Our combined SGD-SVM: 93.58% acc, 6.23% FPR (same ordering of SGD variants).

Appendix B — Feature co-occurrence (paper Fig. 5)

Paper



Ours



Appendix C — Stream bars (S1 vs S2 summary)

